

Intelligence Inside the Circles

How Individual and Organizational Intelligence Levels Function Across Substrate-Indexed Recursive Self-Improvement

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23 August 2026

Abstract

The SARSI-L corpus now contains several distinct notions of “level”: substrate-indexed recursive self-improvement circles, individual intelligence evolution levels, organizational intelligence evolution levels, brain write-depth, authority ceilings, self-state coordinates, and terminal physical bounds. These quantities answer different questions and should not be collapsed into a single ladder. This paper develops a unified account of how **individual intelligence levels I0-I Ω and organizational intelligence levels O0-O Ω function inside each SARSI-L substrate circle**: Circle I software, Circle II hardware, Circle III physical, Circle IV biological, and Circle V+ stellar. The central thesis is that a circle identifies **what substrate an improvement loop reaches**, while I-level identifies **what kind of persistent cognitive evolution an individual can perform**, and O-level identifies **what kind of persistent collective evolution a coordination structure can perform**. Therefore every circle may contain heterogeneous individuals spanning multiple I-levels and organizations spanning multiple O-levels. Higher levels do not replace lower levels; they reorganize their roles. We formalize a general heterogeneous loop consisting of director, discoverer, instrument-maker, specialist, operator, referee, and promoter roles; show how these roles map onto individual evolution and D0-D4 brain write-depth without treating the two scales as equivalent; derive circle-specific role maps; and explain why Circle II and III form a mutual bootstrap rather than a sequence, why Circle IV can progress in parallel, and why Circle V+ increasingly requires organizational rather than singleton intelligence because of the coherence bound on individual minds. The resulting system is best represented as a profile, not a scalar: individual evolution, organizational evolution, substrate-loop completion, brain write-depth, authority, and separation/referee structure must be reported together. The paper concludes with falsifiable transition tests and an experimental program for measuring the interaction among these axes.

Keywords: recursive self-improvement; intelligence levels; organizational intelligence; individual intelligence; multi-agent systems; substrate-indexed RSI; autonomous discovery; open-ended intelligence; SARSI-L; AI agents

1. Introduction

The phrase *level of intelligence* is often used as though it named one ordered quantity. In current AI discourse, however, the phrase can refer to at least four different phenomena: how capable a model is on tasks, how persistently an agent learns, how deeply a system can modify itself, and how widely a group can coordinate. The SARSI-L corpus progressively separates these questions rather than forcing them into one number. Its current Level Map explicitly states that the “level” of an intelligence is a **profile**, and catalogs multiple axes rather than a scalar [6].

The system-scale paper, *Recursive Self-Improvement Is Substrate-Indexed*, introduces a different notion of level: five substrate circles over software, hardware, physical matter, biology, and stellar infrastructure [1]. These circles do not rank cognitive sophistication. They identify the substrate over which the system can close an autonomous improvement loop. Circle I concerns changing software; Circle II concerns changing the hardware substrate; Circle III concerns autonomous manipulation and reproduction in matter; Circle IV concerns biological intervention; Circle V+ concerns self-expanding orbital energy and manufacturing infrastructure.

Separately, the current two-axis framework defines an **Individual Intelligence Evolution Scale I0-I Ω** and an **Organizational Intelligence Evolution Scale O0-O Ω** [7]. I0-I Ω asks what kind of persistent cognitive evolution occurs within one cognitive lineage: response, persistence, adaptive learning, cognitive self-improvement, recursive self-improvement, autonomous discovery, and open-ended cognitive evolution. O0-O Ω asks what kind of persistent collective evolution occurs in a coordination structure: coordination, persistence, adaptation, self-redesign, recursive meta-redesign, collective discovery, and open-ended organizational evolution.

The question of this paper is therefore not “which circle corresponds to I4?” or “is O5 higher than Circle III?” Those questions mix incompatible coordinates. The correct question is:

Within a given substrate circle, what work is performed by individuals at different I-levels, what work is performed by organizations at different O-levels, and how do those levels interact to close or extend the circle?

This paper answers that question by synthesizing the current SARSI-L system, brain, individual, organizational, terminal, Level Map, and two-axis documents [1-7]. It does not replace their definitions. It supplies the missing crosswalk among them.

The core result is simple:

[]

The three coordinates are coupled but not interchangeable.

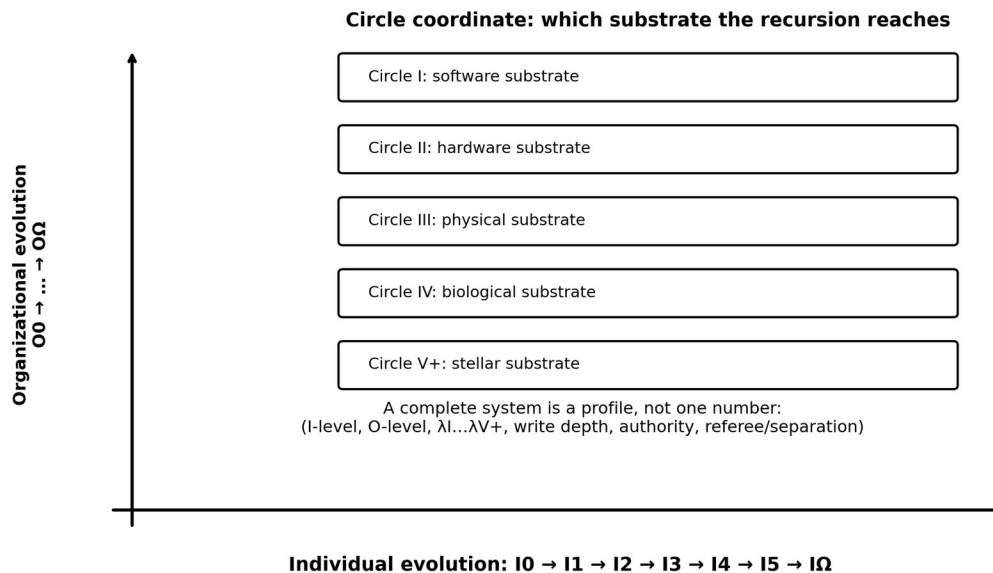


Figure 1. Orthogonal coordinates used in this paper: individual evolution, organizational evolution, and substrate circle.

2. Epistemic Status and Source Discipline

This paper is a **synthesis paper**. Its basic definitions are inherited from the repository rather than newly invented here. In particular:

- Circle definitions and closure criteria are taken from SARSI-L v3 [1].
- Brain write-depth, objective openness, reach, evaluator externality, and the trust inversion are taken from SARSI-B [2].
- The physical definition and bounds of an individual are taken from SARSI-I [3].
- Organizational roles, separation, coverage, and forced heterogeneity are taken from SARSI-O [4].
- The discovery loop, transduction, terminal saturation, and scale-forces-decomposition result are taken from SARSI-Ω [5].
- The non-scalar profile interpretation is taken from the SARSI Level Map [6].

- I0-IΩ, O0-OΩ, the separation gate, the frozen referee for IΩ, and the physical hand-off from I to O are taken from the Two-Axis Framework v2 [7].

Where this paper proposes a mapping across those documents, that mapping is labeled as a **synthesis** rather than a quotation from any single source.

Two anti-inflation rules are especially important throughout. First, memory accumulation is not automatically learning, and learning is not automatically self-improvement [2,7]. Second, organizational behavior above O1 is not a validated organizational level unless proposer, evaluator, and promoter are separated and the evaluator's criteria are outside the proposer's write set [7].

3. The Three Coordinates

3.1 Circle coordinate: which substrate does recursive improvement reach?

SARSI-L defines an improvement loop over domain (d) as:

[L_d = ,, ,]

where the system identifies a candidate improvement, implements it, validates it, and deploys the validated change as the new operative baseline [1]. Closure requires all required steps to execute autonomously in the modal iteration; standing authority to halt can remain external, but required human approval to proceed means the loop remains compensated rather than closed [1].

Table 1. SARSI-L substrate circles

Circle	Substrate	Operational center of the loop	Characteristic limiting floor
I	Software	architecture, training process, weights, software cognition	mainly throughput / compute
II	Hardware	chip specification, design, fabrication, compute integration	mixed informational + physical
III	Physical	machines, manufacturing, operation, repair, self-reproduction	physical construction/material floors
IV	Biological	mechanism discovery, interventions, experiments, clinical deployment	biological + regulatory/physical
V+	Stellar	self-expanding orbital energy/manufacturing infrastructure	physical and orbital floors

A system should not be described as simply “in Circle III.” SARSI-L v3 explicitly rejects binary circle membership and instead reports a completion vector:

[=(I,{II},{III},{IV},_{V+}).]

Several circles may advance simultaneously [1].

3.2 Individual coordinate: what changes in one cognitive lineage?

The Two-Axis Framework v2 retains the following behavioral scale [7].

Table 2. Individual Intelligence Evolution Scale

Level	Name	Defining behavior
I0	Reactive	responds intelligently from current input; no durable evolving cognitive state required
I1	Persistent	durable identity, memory, goals, or task state influence later cognition
I2	Adaptive learning	experience persistently improves knowledge, calibration, strategies, or skills while core learning machinery remains fixed
I3	Self-improving	modifies a cognitive mechanism itself and validates that modification independently
I4	Recursive self-improving	improves the mechanism that generates/evaluates future improvements; improvement competence rises
I5	Autonomous discovery	identifies significant unknowns and produces externally validated new knowledge
IΩ	Open-ended self-evolving	discoveries become cognitive tools that expand the individual's future discovery space; objective revision requires a versioned frozen referee

The I-scale is about **evolutionary behavior**, not one-shot capability. A very capable coding model can remain I0/I1 if its own persistent cognitive mechanism does not validly learn or change.

3.3 Organizational coordinate: what changes in the coordination structure?

Table 3. Organizational Intelligence Evolution Scale

Level	Name	Defining behavior
O0	Coordinated	routes work among members; little durable organizational cognition
O1	Persistent	durable shared memory, roles, task state, evidence, and history
O2	Adaptive	collective history changes allocation, specialist choice, procedure, or knowledge sharing
O3	Self-improving	changes and validates its own routing, communication, memory, verification, or role structure
O4	Recursive self-improving	improves how it discovers, evaluates, promotes, and disseminates future organizational improvements
O5	Autonomous discovery	organization collectively produces externally validated new knowledge with essential distributed cognitive structure
OΩ	Open-ended evolving organization	discoveries repeatedly generate new instruments and organizational forms that expand future collective discovery space

From O2 upward, behavioral evidence is not sufficient unless **separation holds**. If the same locus proposes, judges, and adopts the organizational change, the current framework reports O1 plus an unverified higher-level claim, not a validated O2-O Ω classification [7].

3.4 Brain write-depth is not the same as I-level

SARSI-B defines another axis:

[D0 D1 D2 D3 D4]

where D0 writes no persistent brain state, D1 writes memory, D2 writes policy, D3 writes skills/tools/scaffolds, and D4 writes adapters/weights [2]. This axis describes **where a change lands**, not whether the individual has demonstrated I2, I3, or I4 behavior.

There is therefore no valid identity such as “I3 = D3.” An I3 system may modify a D2 routing policy, a D3 skill architecture, or a D4 model component, provided the change alters its future cognitive mechanism and passes independent evaluation. Conversely, a D4 fine-tuning pipeline operated externally does not make the deployed agent I3.

This distinction is essential for the circle mapping below.

4. The General Heterogeneous Intelligence Loop Inside Any Circle

SARSI-O describes an organization as a set of individuals plus a coordination structure specifying who proposes, evaluates, promotes, holds authority, and couples the system to the world [4]. Its role analysis suggests a general loop that can be reused across every substrate circle.

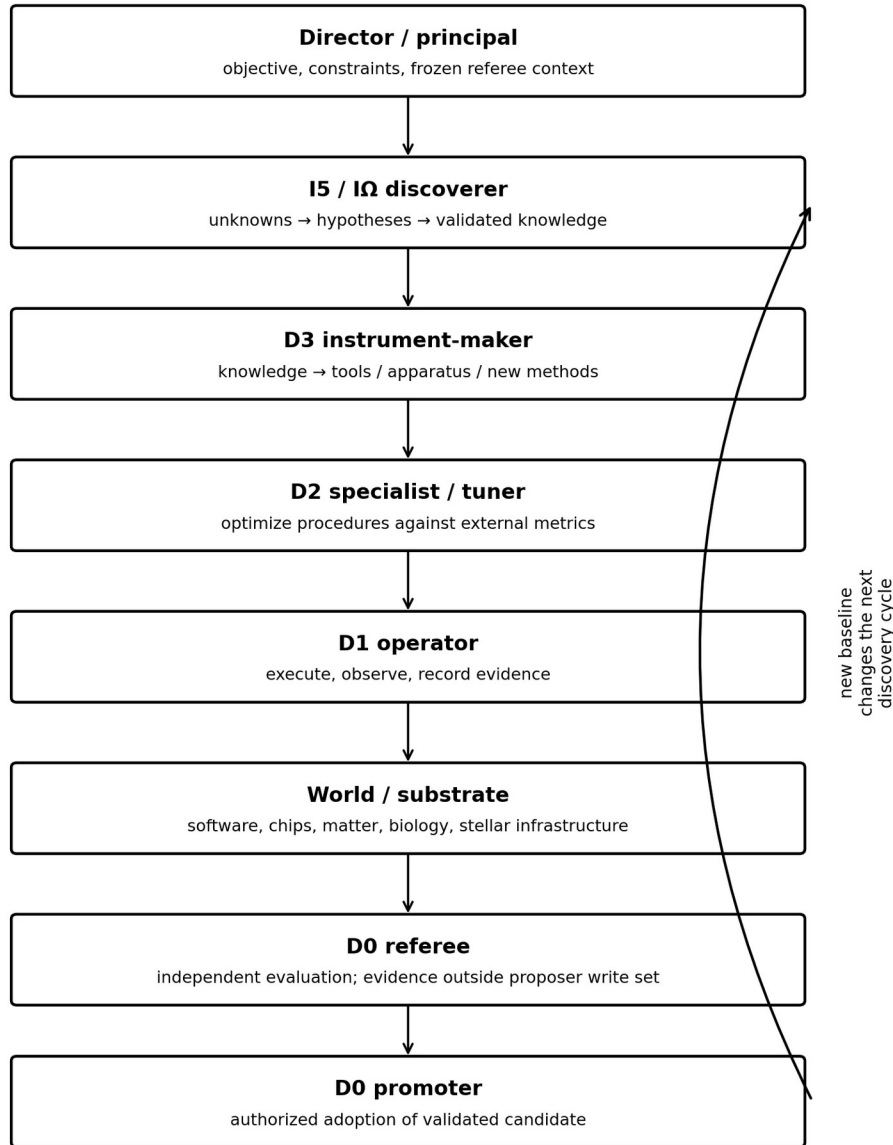
General heterogeneous intelligence loop inside any SARSI-L circle

Figure 2. General heterogeneous intelligence loop inside any SARSI-L circle.

The loop contains seven functional roles:

1. **Director / principal** - defines the current objective, constraints, and admissible frontier. If objectives are revisable, the frozen-referee logic must preserve a pre-revision comparison [2,7].
2. **Discoverer** - at I5/IΩ, identifies important unknowns, formulates hypotheses, and produces candidate new knowledge.
3. **Instrument-maker** - typically associated with D3-capable skill/tool creation; converts knowledge into apparatus, software, methods, or organizational procedures. SARSI-Ω identifies this transduction step as the sole coupling by which knowledge expands reach [5].
4. **Specialist / tuner** - often I2/I3 with D2-D3 write depth; improves procedures against externally supplied metrics.
5. **Operator** - frequently I1/I2; executes procedures, maintains continuity, and produces evidence.

6. **Referee** - commonly D0 with respect to the judged system; evaluates evidence under criteria outside the proposer's write set.
7. **Promoter** - adopts a validated candidate under an authority rule distinct from the proposer.

4.1 Why high and low levels coexist

The organizational paper argues that heterogeneous levels are not merely economically convenient; they are structurally required [4]. A referee must not possess the same effective write reach as the system it judges during the engagement. A proposer and promoter cannot collapse into one locus without invalidating the improvement claim. Faster, shallower loops and slower, deeper loops also operate on different timescales [2,4].

Therefore maturity does not imply that every participant becomes $I\Omega$. In a mature organization:

- high-level individuals create hypotheses, representations, instruments, and meta-improvements;
- mid-level individuals specialize, adapt, and operationalize methods;
- low-level individuals execute mature procedures cheaply and reproducibly;
- restricted D0/D1 individuals or deterministic mechanisms preserve evaluation independence.

The result is **forced heterogeneity** rather than a uniform population of maximal agents [4,7].

4.2 Upward evidence and downward compression

Intelligence flows in two directions.

Upward flow: observation -> persistent episode -> learned pattern -> mechanism problem -> anomaly -> research question -> frontier discovery.

Downward flow: discovery -> representation/tool -> validated procedure -> specialized skill -> routine -> cheap execution.

Bidirectional flow and cognitive compression

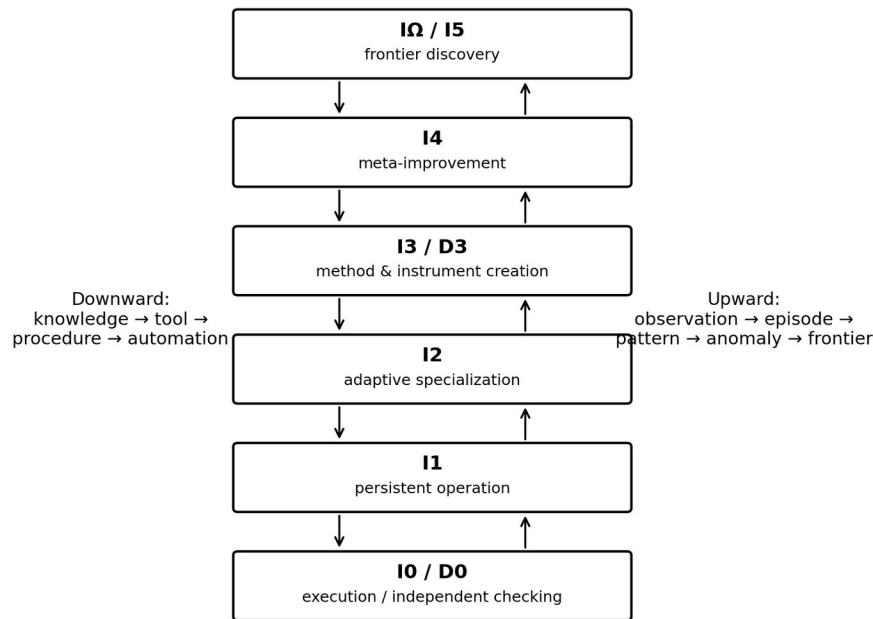


Figure 4. Bidirectional flow and cognitive compression across intelligence levels.

The downward direction is cognitive compression. A difficult problem may initially require an I5/IΩ discovery process; once understood, it may become a D3 tool; once standardized, an I2 specialist can use it; once proceduralized, an I0/I1 executor can run it. Higher intelligence therefore often creates **more work for lower-level intelligence**, not less.

5. Circle I - Software

5.1 Closure target

Circle I is autonomous software self-improvement: identify a modification to the system's architecture or training process, implement it, validate it on a held-out evaluation, and deploy it as the new baseline without required human approval in the modal loop [1]. The crucial point is that *deploy* is part of closure.

5.2 How each individual level functions in Circle I

Table 4. Individual intelligence inside Circle I

I-level	Function in software circle	Typical evidence produced
I0	execute tests, compile code, format artifacts, run fixed benchmarks	deterministic outputs and logs
I1	retain repository state, prior failures, task history, candidate lineage	persistent project evidence
I2	learn which prompts, retrieval strategies, test	held-out future improvement due to learned

I-level	Function in software circle	Typical evidence produced
	sequences, or coding tactics work	state
I3	modify cognitive mechanisms such as retrieval, context gating, planning, tool selection, or memory policy	independently validated mechanism change
I4	improve how candidate cognitive changes are generated, diagnosed, evaluated, or selected	rising improvement competence across generations
I5	discover new algorithms, architectures, training methods, verification techniques, or computational principles	externally validated novel computing knowledge
IΩ	convert discoveries into new cognitive representations/tools that expose previously unreachable software or reasoning problems	repeated frontier expansion under a referee ladder

The decisive transition for Circle I closure is not simply higher benchmark performance. It is the ability to perform the full identify-implement-validate-deploy loop on the operative software substrate.

5.3 How each organizational level functions in Circle I

Table 5. Organizational intelligence inside Circle I

O-level	Function in software circle
O0	routes coding, testing, review, and execution tasks among agents
O1	preserves shared repository history, candidate lineage, evidence, roles, and unresolved issues
O2	learns which agents/workflows are effective for which software changes and updates allocation accordingly
O3	redesigns and validates its own software-development routing, memory, review, testing, or verification structure
O4	improves the meta-process that generates and evaluates future software-organization redesigns
O5	collectively discovers new computing principles or engineering methods no single member independently contains
OΩ	discoveries repeatedly create new software research instruments and new organizational forms, expanding the collective software-discovery frontier

5.4 A valid Circle-I recursive organization

A mature Circle-I organization can contain an I4 proposer, an independent D0 referee, a distinct promoter, I2 specialist agents, I1 persistent operators, and I0 deterministic test executors. This is not a contradiction. It is the architecture that makes the I4 claim meaningful.

A closed Circle-I loop does **not** require removing all governance. It requires removing *required human approval from the modal loop*. Separation can be preserved by autonomous external evaluators/promoters whose criteria the proposer cannot rewrite. This is the correct reconciliation between SARSI-L closure and SARSI-B/O separation.

6. Circle II - Hardware

6.1 Closure target

Circle II requires autonomous chip specification, production-ready physical design, fabrication at viable yield, and integration of the resulting hardware into the system's own compute infrastructure [1]. SARSI-L v3 emphasizes that chip **design** and chip **manufacture** are different problems: the former is largely informational; the latter enters physical infrastructure.

6.2 How each individual level functions in Circle II

Table 6. Individual intelligence inside Circle II

I-level	Function in hardware circle
I0	run EDA tools, fixed simulation, verification, layout checks, characterization scripts
I1	preserve design histories, fab results, yield failures, process versions, integration lineage
I2	learn reusable design heuristics, process tuning strategies, verification priorities, yield predictors
I3	improve chip-design/search/verification mechanisms or hardware-aware cognitive tools
I4	improve the process that discovers and validates future hardware-design improvements
I5	discover new architectures, device principles, materials relationships, packaging or fabrication concepts
I Ω	turn new hardware knowledge into new design/fabrication instruments that expose previously unreachable hardware search spaces

An I Ω individual still does not close Circle II by thinking. Fabrication is outside the individual cognitive substrate. This is exactly why substrate indexing matters: cognitive reach and physical reach can diverge sharply.

6.3 How each organizational level functions in Circle II

Table 7. Organizational intelligence inside Circle II

O-level	Function in hardware circle
O0	coordinates architecture, design, simulation, verification, fab, and integration tasks
O1	preserves shared design/fab evidence, process histories, versioned specifications, and failure records
O2	learns allocation between design specialists, process nodes, tools, and verification strategies
O3	redesigns and validates its own hardware-development and qualification workflow
O4	improves how it discovers better hardware-development organizations and validation processes

O-level	Function in hardware circle
O5	collectively discovers new hardware/device/manufacturing knowledge
O Ω	converts discovery into new fabrication tools and organizational forms that continuously expand hardware reach

The binding organizational challenge increasingly shifts from cognition to integration with external physical infrastructure.

7. Circle III - Physical Matter and Self-Reproduction

7.1 Closure target

Circle III requires the system to identify a physical need, design machines/processes, manufacture them from available materials, operate them under unforeseen failures, and repair or replace components using in-situ resources [1]. The relevant criterion is not “useful robot.” It is a substantially self-maintaining and self-reproducing physical loop.

7.2 How each individual level functions in Circle III

Table 8. Individual intelligence inside Circle III

I-level	Function in physical circle
I0	execute motor/control subroutines, inspection steps, fixed machining/manufacturing procedures
I1	preserve machine state, maintenance history, resource inventories, failure episodes
I2	adapt operation, maintenance, scheduling, local control, process parameters
I3	improve control architectures, manufacturing plans, diagnostic mechanisms, or robotic skill structures
I4	improve the process that generates and validates future physical-system improvements
I5	discover new materials, mechanisms, manufacturing principles, failure physics, or resource pathways
I Ω	transform discoveries into new physical instruments that make previously impossible forms of manipulation reachable

In Circle III, D3 instrument-making becomes especially concrete: the output is not only a software skill but a robot, machine tool, process, sensor, material workflow, or manufacturing cell.

7.3 How each organizational level functions in Circle III

Table 9. Organizational intelligence inside Circle III

O-level	Function in physical circle
O0	coordinates machines, robots, manufacturing cells, logistics, and inspection
O1	preserves shared production state, maintenance history, resource maps, machine lineage
O2	learns allocation, scheduling, maintenance, routing, and specialization from production history
O3	redesigns its own manufacturing topology, verification, maintenance, logistics, or role structure
O4	improves the process that discovers and validates future production-system redesigns
O5	collectively discovers new physical mechanisms/material/process knowledge
O Ω	discoveries generate new production technologies and new organizational structures that recursively expand physical reach

7.4 The matter bottleneck

Higher I-levels can reduce cognitive latency dramatically while Circle III remains constrained by construction, transport, materials, qualification, and failure physics. SARSI-L calls this the **matter bottleneck**: after cognitive throughput accelerates, physical-process floors dominate [1].

This creates an important asymmetry: I4/I5/I Ω may increase the *quality* of designs rapidly, while ($_III$) changes slowly because the world must still execute physical transitions.

8. Circle II and Circle III Form a Mutual Bootstrap

Earlier versions treated the circles as a simple sequence. SARSI-L v3 replaces that picture with a dependency graph in which Circle II hardware and Circle III physical autonomy are partially mutually dependent [1]. Autonomous fabs require advanced physical automation; self-reproducing physical systems require electronics manufacturing. Neither must be fully closed before the other advances.

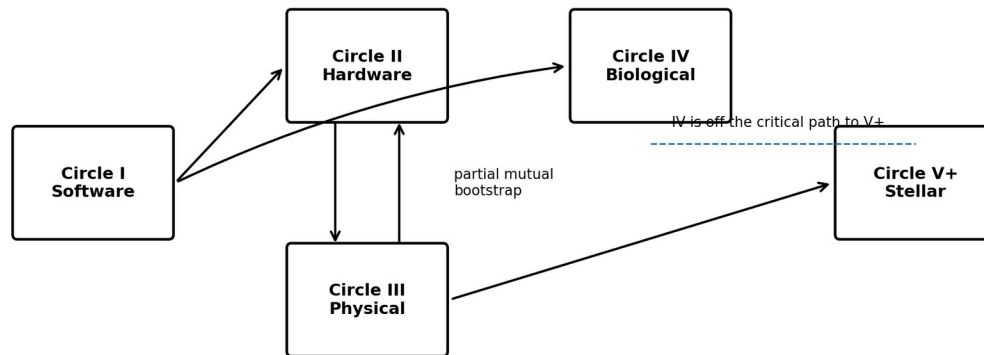
SARSI-L v3 dependency structure: not a simple chain

Figure 3. Cross-circle dependency structure in SARSI-L v3.

The resulting bootstrap is:

$[\{II\}^{\wedge(1)} \{III\}^{\wedge(1)} \{II\}^{\wedge(2)} \{III\}^{\wedge(2)}]$

This has a direct intelligence-level interpretation. A Circle-II I3 hardware-tool improvement may enable a Circle-III I2/I3 physical automation improvement, which enables better fab operation, which returns to Circle II. The recursive unit is therefore not always one individual or one organization. It can be a **cross-circle coupled system** whose local improvements alternate between substrates.

Organizationally, this suggests a coupled pair of O2-O4 structures sharing evidence but preserving role separation: a hardware organization and a physical-manufacturing organization whose improvements are mutually enabling.

9. Circle IV - Biological Discovery and Intervention

9.1 Closure target

Circle IV requires autonomous biological hypothesis generation and testing, intervention design, validation through experimental and clinical pathways, population-scale deployment, monitoring, and iteration [1]. It contains physical floors that cognition cannot simply compress, including organism response and clinical endpoints.

9.2 How each individual level functions in Circle IV

Table 10. Individual intelligence inside Circle IV

I-level	Function in biological circle
I0	execute fixed assays, image analysis, preprocessing, statistical routines, protocol steps
I1	retain experiment, cohort, protocol, failure, and evidence histories
I2	learn better experiment selection, parameter tuning, prioritization, or analysis strategies

I-level	Function in biological circle
I3	improve scientific reasoning, causal inference, experimental-design, retrieval, or simulation mechanisms
I4	improve the process by which future scientific-method improvements are generated and evaluated
I5	autonomously discover and validate previously unknown biological mechanisms or relations
IΩ	transform discoveries into new assays, representations, experimental methods, or cognitive tools that expose deeper biological unknowns

Circle IV is the domain in which I5 maps most directly onto the substrate loop because autonomous discovery is explicitly part of the circle's operational criterion.

9.3 How each organizational level functions in Circle IV

Table 11. Organizational intelligence inside Circle IV

O-level	Function in biological circle
O0	coordinates literature, theory, experiment, assay, analysis, replication, and reporting
O1	preserves shared evidence, provenance, protocols, biological models, and unresolved contradictions
O2	learns better allocation of experiments, specialists, modalities, cohorts, and verification resources
O3	redesigns and validates its own research workflow, replication architecture, memory, review, or laboratory organization
O4	improves how it discovers and evaluates better research organizations and scientific methods
O5	collectively identifies unknowns and produces validated biological knowledge no single member holds alone
OΩ	discoveries repeatedly generate new instruments and scientific organizations that expand collective biological discovery space

O5 is especially natural in Circle IV because no single individual need contain all essential reasoning. One member may formulate a mechanism, another design a discriminating experiment, another build an assay, another perform independent analysis, and another referee the claim. Organizational cognition becomes constitutive of the discovery rather than mere parallelism.

9.4 Circle IV is partly parallel to the hardware-physical critical path

SARSI-L v3 explicitly removes the earlier assumption that biology lies on the path to stellar engineering. Circle IV requires advanced software cognition but is not a prerequisite for Circle V+ [1]. This means a mature system may show:

[{IV} {III}]

or the reverse. Circle status is therefore a vector, not a stage number.

10. Circle V+ - Stellar-Scale Infrastructure

10.1 Closure target

Circle V+ is not merely orbital energy collection. The criterion is a self-expanding orbital manufacturing loop that designs, constructs, and operates collectors and uses the captured energy to produce further net-positive growth of productive collection area [1].

10.2 How each individual level functions in Circle V+

Table 12. Individual intelligence inside Circle V+

I-level	Function in stellar circle
I0	execute local control, navigation, inspection, maintenance, telemetry, and fixed manufacturing tasks
I1	preserve local infrastructure state, mission history, resource maps, and failure lineage
I2	adapt local scheduling, maintenance, navigation, allocation, and production strategies
I3	improve local engineering cognition, control, manufacturing, or planning mechanisms
I4	improve the process generating future local infrastructure/cognitive improvements
I5	discover new physical, materials, orbital, energy, or engineering principles
I Ω	locally transform discoveries into new instruments and cognitive methods that expand the frontier accessible to that individual

The qualification *locally* becomes important. SARSI-I derives a coherence bound for a singleton mind:

[R ,]

so a coherent individual cannot grow arbitrarily large at fixed decision speed [3,7]. At stellar scales, the intelligence architecture must increasingly decompose into multiple individuals exchanging delayed messages.

10.3 How each organizational level functions in Circle V+

Table 13. Organizational intelligence inside Circle V+

O-level	Function in stellar circle
O0	coordinates local missions/nodes
O1	maintains persistent distributed infrastructure records and role identities
O2	adapts resource allocation, specialization, scheduling, replication, and communication from collective history
O3	redesigns its distributed coordination architecture, routing, verification, resource allocation, or institutional topology
O4	improves how it discovers/evaluates future distributed organizational redesigns under communication delay

O-level	Function in stellar circle
O5	collectively produces new physical/engineering knowledge across distributed regions
OΩ	discoveries generate new instruments and new distributed organizational forms, enabling open-ended collective frontier expansion despite the impossibility of one global workspace

At Circle V+, O-level becomes increasingly important because physical scale forces pluralism. SARSI-Ω's scale-forces-decomposition result says there is no singleton at the top; stellar-scale cognition is necessarily federated [5]. The individual axis does not continue upward forever. It hands off to organizational development [7].

11. The Complete Circle × Individual Matrix

The following matrix summarizes the dominant function of each individual evolutionary level within each substrate circle. It is a synthesis, not an assertion that every implementation must use exactly these roles.

Individual intelligence levels across SARSI-L substrate circles

I-level	Circle I Software	Circle II Hardware	Circle III Physical	Circle IV Biological	Circle V+ Stellar
I0	execute / tests	EDA / simulation	control / inspection	assays / analysis	local execution
I1	project memory	design / fab memory	machine history	experiment / cohort memory	local infrastructure history
I2	adaptive coding / retrieval	adaptive design / process	adaptive operation	adaptive experiment strategy	adaptive local operations
I3	improve cognitive / software methods	improve design tools / methods	improve control / manufacture methods	improve scientific methods	improve local engineering cognition
I4	improve improvement process	improve hardware-improvement process	improve physical-improvement process	improve science-improvement process	improve local infrastructure-improvement process
I5	discover computing principles	discover hardware / device principles	discover materials / mechanisms	discover biological mechanisms	discover stellar / physical principles
IΩ	cognitive tools expand software frontier	tools expand hardware frontier	tools expand physical frontier	instruments expand biological frontier	local open-ended frontier expansion; no single stellar n

Figure 5. Individual intelligence levels across SARSI-L substrate circles.

Two patterns are visible. First, I0-I4 increasingly improve the **machinery of operation and improvement**. Second, I5-IΩ increasingly expand the **frontier of valid knowledge and reachable problems**.

12. The Complete Circle × Organizational Matrix

Organizational intelligence levels across SARSI-L substrate circles

O-level	Circle I Software	Circle II Hardware	Circle III Physical	Circle IV Biological	Circle V+ Stellar
00	coordinate coding	coordinate design / fab	coordinate production	coordinate research	coordinate missions
01	shared repo / evidence	shared design / fab state	shared production state	shared scientific evidence	distributed infrastructure memory
02	learn task allocation	learn design / fab allocation	learn production allocation	learn research allocation	learn distributed resource allocation
03	redesign dev organization	redesign hardware workflow	redesign manufacturing organization	redesign research organization	redesign distributed coordination
04	meta-improve organizational improvement	meta-improve hardware organization	meta-improve production organization	meta-improve research organization	meta-improve distributed organization
05	collective computing discovery	collective hardware discovery	collective physical discovery	collective scientific discovery	collective stellar engineering discovery
0Ω	open-ended software research organization	open-ended hardware / fab organization	open-ended physical production organization	open-ended scientific organization	open-ended distributed civilization-scale organization

Figure 6. Organizational intelligence levels across SARSI-L substrate circles.

Again, $O\Omega$ is not Circle V+. An $O\Omega$ organization could exist while physical circles remain incomplete, and a civilization could possess enormous physical resources while its organizational intelligence remains low. The Level Map makes the same point for energy: high energy capture is an indicator, never a definition of intelligence [5,6].

13. A Unified State Representation

The repository's current direction suggests that no single label can fully describe an advanced system. A minimally adequate profile is:

[=]

where:

- (I) = individual evolution level or profile;
- (O) = organizational behavioral level plus separation status;
- () = completion across Circle I-V+;
- (D) = brain write depth;
- (A) = authority ceiling;
- (R) = self/other reach;
- (G_O) = organizational structure profile: coverage, separation, throughput, referee externality, world coupling;
- () = physical-floor saturation per domain;
- (E) = captured/usable energy, an indicator rather than a constitutive intelligence measure.

This expression is not intended as a scalar score. It is a reporting schema.

13.1 Why a scalar fails

A system can be high on one coordinate and low on another:

- I4 but O1: powerful self-improving individual in a weak organization.
- I2 but O4: modest individuals embedded in a sophisticated meta-improving institution.
- High (I), low (O): software recursion with a matter bottleneck.
- High energy, low O-separation: large physical capability with unreliable organizational truth formation.
- I Ω locally, O2 globally: open-ended individuals whose collective structure cannot yet evolve open-endedly.

These are qualitatively different systems even if a benchmark reports similar task performance.

14. Coexistence Is a Structural Requirement

14.1 Separation forces heterogeneous effective levels

From O2 upward, proposer, evaluator, and promoter must be distinct [7]. The evaluator must not share the proposer's effective write set. Therefore even an organization populated by identically capable maximal individuals must restrict their roles differently during any valid improvement decision. In effect, the organization must create heterogeneity.

14.2 Timescale forces heterogeneous write depths

SARSI-B argues that write depth corresponds to different timescales and risks [2]. Memory changes can occur frequently; policy changes less frequently; new skills/tools require stronger evaluation; weight changes are slower and more consequential. An organization that forced every operation through D4-like governance would be unusably slow, while one that treated D4 changes like D1 memory updates would be unsafe and empirically uninterpretable.

14.3 Physical bounds force heterogeneous individual designs

SARSI-I shows that individual radius, speed, and workspace width trade against one another [3]. A fast narrow individual and a slower wider individual may occupy different optimal points on the physical frontier. Therefore heterogeneity can remain rational even if computational resources become abundant.

14.4 Cognitive compression makes lower levels more valuable over time

Once a high-level discovery becomes a mature instrument or procedure, lower-level agents can execute it at high scale and reliability. Thus mature systems may contain **more I0/I1 execution** precisely because their I5/I Ω components have succeeded at converting difficult cognition into routine machinery.

15. Interaction Between Individual and Organizational Recursion

There are at least three recursive loops in the combined framework.

15.1 Individual recursion

[I_t I_{t+1}.]

At I4 the improvement mechanism itself improves. At IΩ discoveries can create new cognitive tools that expand future discovery space [7].

15.2 Organizational recursion

[O_t O_{t+1}.]

At O4 the mechanism for generating and evaluating organizational redesign improves. At OΩ discoveries create new organizational forms that expand collective discovery space [7].

15.3 Substrate recursion

[_d d .]

Cross-circle coupling allows one substrate improvement to accelerate another. Better hardware accelerates software experimentation; better physical automation improves fabrication; better instruments improve biological discovery; larger energy infrastructure expands the feasible experiment/manufacturing budget [1,5].

15.4 Coupled recursion

The complete system therefore behaves approximately as:

[]

This coupled loop is the main synthesis of this paper.

16. What Changes as the Circle Moves Outward?

The same intelligence hierarchy can operate in every circle, but the **binding floor changes**.

16.1 Circle I: cognition is near the substrate

Most loop operations can be expressed as computation. Therefore I3/I4 and O3/O4 improvements can directly shorten loop time.

16.2 Circle II: cognition meets manufacturing

Design can accelerate quickly; fabrication cannot. Organizational coupling to fabs becomes critical.

16.3 Circle III: physical process dominates

Even excellent cognition must wait for construction, material transformation, wear, testing, repair, and replication. Parallelism rather than reasoning speed becomes central.

16.4 Circle IV: discovery and physical time coexist

Scientific cognition may accelerate hypothesis generation, but organism response and clinical endpoints remain physical floors.

16.5 Circle V+: communication and scale force organization

Large distances make one coherent workspace physically impossible at useful cycle times. Organizational intelligence becomes the dominant path for continued scaling.

This is why the same I-level means different practical leverage in different circles. I4 may radically change Circle-I iteration speed and have much less effect on a five-year biological endpoint.

17. Current AI Systems as an Illustrative Boundary Case

The current Two-Axis Framework classifies common coding-agent systems primarily around I1 and O0-O1 unless longitudinal learning or separated organizational improvement is demonstrated [7]. This paper does not reclassify those systems. It draws one implication from that placement: **current agent systems are concentrated in the early part of Circle I**, and much of what appears as recursive capability is still a compensated loop in which external humans or organizations supply missing identify, evaluation, authorization, or deployment functions [1,7].

This matters because adding more agents does not automatically move a system outward in circle space or upward in O-level. Ten I1 agents coordinated in parallel can still be O1, and an O3-like redesign pipeline that self-grades without separation remains O1 plus an unverified O3 claim [7]. Similarly, a coding agent that writes new software is not I3 unless its own future cognition actually runs on the changed mechanism and the change passes independent evaluation.

18. Falsifiable Transition Tests for the Combined Framework

A cross-scale framework should produce measurements, not only diagrams. The following tests are proposed as a synthesis of the repository's existing anti-inflation rules.

Table 16. Tests of individual level within a circle

Claimed transition	Required evidence inside a circle
I0 -> I1	persistent state from earlier circle work is recovered and correctly affects later work
I1 -> I2	matched held-out future circle performance improves because of learned state, not replayed instructions
I2 -> I3	a cognitive mechanism relevant to the circle changes and independently improves held-out outcomes
I3 -> I4	the process generating/evaluating improvements itself shows rising validated improvement competence
I4 -> I5	the individual produces externally validated novel knowledge relevant to the circle
I5 -> I ∞	discoveries repeatedly create cognitive tools enabling previously inaccessible discovery classes, with a retained referee ladder

Table 17. Tests of organizational level within a circle

Claimed transition	Required evidence inside a circle
O0 -> O1	durable shared role/task/evidence state survives sessions/member turnover
O1 -> O2	organizational history causes validated improvement in allocation/procedure on held-out work, with separation
O2 -> O3	the organization changes a coordination mechanism and independently validates the change
O3 -> O4	the mechanism generating/evaluating organizational changes itself becomes better under independent evaluation
O4 -> O5	distributed organization produces externally validated novel knowledge requiring essential collective structure
O5 -> O Ω	discoveries repeatedly create new instruments/organizational forms that expand collective discovery space

18.1 Circle completion remains a separate test

Even if an organization contains I Ω and O Ω behavior, Circle (d) is not closed until the domain-specific identify-implement-validate-deploy criterion is met. Conversely, a narrow highly automated domain loop could approach closure without containing I Ω .

This distinction is one of the strongest reasons to preserve the three-coordinate model.

19. Experimental Program

A practical research program can test the interaction among I, O, and Circle coordinates.

E1 - Same I-level, different circles

Hold one agent architecture fixed and deploy it in software, hardware-design, robotic-control, and biological-design tasks. Measure how much its I2/I3 improvements reduce end-to-end loop period in each domain. Prediction: gains fall as physical floors dominate.

E2 - Same circle, different I-level compositions

Within Circle I, compare organizations dominated by I1 agents, mixed I1-I3 agents, and mixed I1-I4 agents while holding total compute fixed. Measure task gain, improvement competence, evaluation cost, and regression.

E3 - Same individuals, different O-level structure

Hold member agents fixed and compare O1 coordination, O2 adaptive allocation, and O3 self-redesign under identical held-out workloads. This directly tests whether organizational intelligence is structural rather than a proxy for member capability.

E4 - Separation ablation

Compare a redesign pipeline where proposer=evaluator=promoter to one with strict separation. Measure false promotion, held-out regression, and apparent versus externally verified improvement. The framework predicts that unseparated systems will systematically overstate improvement.

E5 - Cognitive compression

Track a problem from first I5 discovery through D3 instrument creation, I2 specialist operation, and eventual I0/I1 routine execution. Measure the decline in cognitive cost and the increase in throughput after proceduralization.

E6 - Circle II/III bootstrap

Measure alternating improvements in chip/fab automation and physical-manufacturing automation. Test whether progress takes the predicted ladder form rather than strict sequential closure.

E7 - Biological physical floor

Increase I-level/O-level cognition while holding biological endpoint time fixed. Prediction: hypothesis throughput rises substantially while some endpoint latency remains unchanged.

E8 - Individual-to-organization hand-off

Increase the simulated physical distribution of a single decision system until communication delay violates the coherence-cycle requirement; compare performance to a decomposed multi-individual organization. This operationalizes the I-to-O hand-off proposed in SARS-I and Two-Axis v2 [3,7].

E9 - O Ω without outer-circle closure

Construct an open-ended scientific organization in a software-only or simulation-rich domain. If it repeatedly creates new instruments and organizational forms while ($_ \{III\}$) remains low, this demonstrates that O Ω and Circle V+ are not equivalent.

E10 - High-circle capability with low O-level

Construct or simulate a physically capable infrastructure with static organization. If it controls large resources yet fails to adapt or validate organizational change, this demonstrates that physical reach/energy and O-level are separable.

20. Implications for AI4Science and Research-Agent Design

The combined framework suggests that an AI4Science platform should not aim to make every agent maximal. It should aim to **cover the required roles with the appropriate effective intelligence, write depth, authority, and separation.**

A research organization might therefore include:

- I5/I Ω research agents for unknown discovery and representation invention;
- I3/I4 method agents for improving scientific cognition and the improvement process;
- D3 instrument agents for turning knowledge into assays, software, simulations, or experimental tools;
- I2 domain specialists for adaptive scientific operations;
- I1 persistent operators for longitudinal task continuity;
- I0 deterministic executors for reproducible routine work;
- D0 or externally controlled referees/promoters for evaluation and promotion.

The circle then specifies which substrate this organization is acting upon. A drug-design research organization may primarily inhabit Circle IV while relying on Circle I software infrastructure and Circle II/III experimental hardware. A robotics research organization may straddle Circle I and III. A hardware research organization may live in the II->III bootstrap. The organization should therefore report a **circle-completion vector**, not one circle label.

21. Limitations

First, the I Ω -I Ω and O Ω -O Ω scales are conceptual taxonomies rather than validated psychometric measures. Their transition tests are designed to resist inflation but have not been standardized across the field [7].

Second, this paper's role mapping is a synthesis. SARSI-O maps organizational roles primarily with brain write-depth/open/reach notation, not directly with I Ω -I Ω . The mapping here therefore describes likely functional correspondence, not a theorem that a given role requires one exact I-level.

Third, the physical coherence argument defines an individual by a single global workspace. A fundamentally different account of individuality could change the I-to-O hand-off [3,7].

Fourth, O-level separation is treated largely as a gate. Real organizations may have partial independence, correlated evaluators, shared training distributions, or indirect influence that make binary separation too coarse [7].

Fifth, circle completion weights remain estimated rather than measured. SARSI-L itself emphasizes that (d) should currently be treated as an ordering rather than a precision measurement [1].

Sixth, open-ended I Ω /O Ω behavior may be difficult to distinguish from drift unless referee ladders and external evidence are retained. Pairwise non-regression also does not by itself bound long-run objective drift [7].

Finally, the framework deliberately separates intelligence from desirability. A highly adaptive individual or organization can pursue poor goals. Higher I, O, or circle completion does not by itself imply beneficial outcomes.

22. Conclusion

The SARSI-L framework is easiest to understand once its several uses of "level" are separated.

A **circle** asks:

What substrate can the recursive loop autonomously improve?

An **individual level** asks:

How does one cognitive lineage persist, learn, self-improve, recursively improve, discover, and evolve?

An **organizational level** asks:

How does a coordination structure persist, learn, redesign itself, recursively improve its redesign process, discover collectively, and evolve open-endedly?

A **brain write-depth** asks:

How deeply can a particular improvement write into persistent cognition?

These are different coordinates of one system.

Inside every circle, multiple individual levels and organizational roles coexist. I0/I1 components execute and preserve; I2 adapts; I3 improves mechanisms; I4 improves the improvement process; I5 discovers; I Ω turns discoveries into new cognitive reach. O0 coordinates; O1 preserves collective state; O2 adapts allocation; O3 redesigns organization; O4 improves organizational improvement; O5 discovers collectively; O Ω converts discovery into new collective forms and new reach. Independent referees and promoters remain structurally distinct because improvement claims cease to be well-posed when the same write locus judges itself.

As the circles move outward, the cognitive architecture may remain similar while the rate-limiting substrate changes. Circle I is close to computation. Circle II reaches fabrication. Circle III encounters the matter bottleneck. Circle IV combines scientific acceleration with biological time. Circle V+ forces decomposition into organizations because no single coherent workspace can span stellar distances at useful cycle times.

The resulting picture is not a ladder but a nested, coupled set of recursive loops:

[]

or, in operational form:

[]

The most mature intelligence is therefore not defined by having only maximal individuals, one maximal organization, or one completed circle. It is defined by **the structure that keeps all of these loops validly coupled while preserving the distinctions that make improvement measurable at all.**

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Repository basis

The primary source repository for this synthesis is:

<https://github.com/integritynoble/sarsi-intelligence-level>